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Does Political Decoupling Through Fair Market Privatization Improve Managerial and Political Outcomes? Insights From Water Utility Sales

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ABSTRACT

Many U.S. municipalities face environmental service delivery challenges, including aging infrastructure and rising debt. In response, several states have adopted Fair Market Value (FMV) legislation, which aims to incentivize private firms to acquire and invest in municipal utilities—particularly in the water sector. However, the impacts of FMV privatization remain understudied. Guided by political decoupling theory, this study hypothesizes that FMV privatization improves drinking water utility management and local incumbent electoral outcomes. Using a difference-in-differences design, we analyze fifteen Illinois municipalities that sold their utilities after the state adopted FMV legislation relative to thirty-four controls. Results suggest that FMV privatization improves personnel and technical management outcomes by reducing violations under the Safe Drinking Water Act but has no significant impact on incumbent vote share. These findings indicate that while FMV privatization may enhance utility management through political decoupling, its presumed political benefits warrant further investigation.

1 | Introduction

In the 21st century, many municipalities across the United States and globally are struggling to deliver environmental public services like drinking water provision, thanks to multiple stressors including aging infrastructure, climate-related threats, regulatory compliance challenges, and rising debt (Deslatte and Stokan 2020; Garcia et al. 2019; Hughes et al. 2020; Koppenjan and Enserink 2009). In the U.S., addressing these issues can be both financially costly and politically risky (Mullin and Hansen 2023; Teodoro et al. 2020); consequently, many local elected officials have refrained from doing so (Hansen and Mullin 2022). In response, numerous states have adopted Fair Market Value (FMV) legislation to encourage privatization of small and distressed municipal utilities, particularly water utilities, with the goal of driving consolidation and capital investment (GAO 2021). Importantly, these laws target a particular

kind of privatization, divestiture, in which a municipality sells both utility operations and infrastructure outright to a private buyer, rather than other common forms of privatization, such as concession contracts.

FMV legislation reshapes incentives for both municipal sellers and investor-owned utility (IOU) buyers by replacing traditional book value methodology (i.e., original utility construction cost minus depreciation) with independently appraised market valuation (i.e., FMV) as the basis for utility divestiture and subsequent rate base adjustments (McKiernan and Warner 2024). Under these laws, municipalities can voluntarily sell their utilities to large IOUs at “fair market” prices that are inflation-adjusted (subject to public utility commission approval), often resulting in substantially higher sale prices. FMV-authorized sales enable IOUs to include both the FMV acquisition price and subsequent capital investments in the rate base on which they

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Evidence for Practice

- Municipal policymakers and utility managers should assess whether their state has adopted Fair Market Value (FMV) legislation and evaluate whether an FMV utility sale could help achieve local infrastructure and fiscal goals.
- FMV privatization may enable municipalities to secure higher sale prices compared to traditional book value appraisals, generating revenue that can support other local priorities.
- FMV privatization is associated with improvements in drinking water utility performance, particularly regarding compliance with personnel and technical requirements under the Safe Drinking Water Act.
- Although FMV privatization may improve utility management outcomes, it offers no clear electoral benefit for local incumbents and may even result in political costs.
- Although FMV privatization may offer municipalities higher sale prices and improved short-term utility performance, these benefits come with trade-offs, including rate increases and uncertain impacts on affordability and other longer-term goals.

earn regulated returns. Therefore, rather than being limited to earning returns only on depreciated book values, IOUs can earn returns on the full FMV purchase price and subsequent capital investments. This creates stronger financial incentives for IOUs to acquire and invest in distressed systems while also providing municipalities with substantial financial windfalls (McKiernan and Warner 2024; Warner 2024).

California was the first state to adopt FMV legislation in 1997, and since 2013, twelve additional states have adopted similar legislation (Kline 2021; McKiernan and Warner 2024). Despite the growing prevalence of state FMV laws and the subsequent increase in utility acquisitions (Beecher 2019; GAO 2021; McKiernan and Warner 2024), no scholarship has examined the local impacts of this emerging form of public service privatization.

Guided by the political decoupling theory (Teodoro et al. 2020), we hypothesize that FMV privatization of municipal drinking water utilities improves both managerial and political outcomes at the local level. We test these propositions using a quasi-experimental difference-in-differences research design. Our treatment group includes fifteen Illinois municipalities that sold their drinking water utilities between 2013, when the state adopted FMV legislation, and 2023. We matched those cases with thirty-four descriptively similar control municipalities that maintained utility ownership and are located in the same counties. We measure management outcomes using Safe Drinking Water Act (SDWA) utility compliance data (i.e., personnel and technical violations) and measure political outcomes using incumbent municipal legislators' (e.g., city councilors, village trustees) election vote share data.

The managerial results indicate that FMV privatization is associated with significant improvements in both utility personnel

management (i.e., reductions in SDWA monitoring and reporting violations) and technical management (i.e., reductions in SDWA health violations). By contrast, although we expected FMV privatization to increase incumbent vote share by shifting responsibility for utility decisions away from local officials (Teodoro et al. 2020; Walker et al. 2025) and increasing funds for other municipal priorities, the political effects of privatization are insignificant. Their negative directionality may even suggest that privatization carries some political risks, rather than the anticipated electoral benefits. However, given the small sample size available for the political analysis, future studies using larger, multi-state datasets are needed to better understand if and how FMV privatization affects local political outcomes.

Taken together, these findings offer quasi-experimental empirical evidence on the effects of FMV-related utility privatization. The management results lend support to FMV legislation and subsequent utility sales, but the electoral results raise important questions about its assumed political advantages. As more states adopt FMV legislation and more municipalities voluntarily privatize their utilities following the FMV regulatory framework, it will be prudent to understand the long-term implications of these decisions. Future research should explore the effects of FMV privatization in other contexts, including factors not assessed in this study (e.g., service rates and affordability, book value versus FMV privatization), to assess the generalizability of our results and whether FMV privatization is a sustainable solution to infrastructure governance challenges.

This article begins by reviewing scholarship on the limitations of privatization in addressing local public service challenges and assesses whether FMV privatization offers a more effective alternative. We then apply the theory of political decoupling to develop hypotheses about its effects on management outcomes for utilities and electoral outcomes for municipal elected officials. We test these hypotheses empirically using a difference-in-differences research design and conclude with a discussion of the implications of our findings, the limitations and scope conditions of our results, and opportunities for future research.

2 | Public Service Privatization

Public administration scholars have assessed the causes and effects of local service privatization (i.e., service delivery divestiture to investor-owned firms) for decades. The New Public Management (NPM) era in the 1980s led to a rapid increase in public service privatization at the local level (Alonso et al. 2015; Bel and Fageda 2017). Scholars and practitioners alike argued that privatization would inject market-based competition and economy-of-scale efficiencies into service delivery, increasing quality and reducing costs (Warner and Hebdon 2001). As a result, the rate of local public service privatization increased throughout the 1980s and 1990s (Warner and Hefetz 2008).

However, the inherently monopolistic nature of the utility sector undermines many of the competition-based assumptions of the NPM framework. For example, residents without private wells cannot choose a different provider unless they relocate. This lack of competition may explain why many privatized utilities, particularly water utilities, have failed to meet community goals

(e.g., updated infrastructure, reduced costs, improved service quality) (Bakker 2013; Girth et al. 2012; Post and Ray 2020; Warner 2024). While the evidence of market forces improving water quality is scant, evidence of their effects on investment and revenue is not. As investor-owned utilities are legally obligated to generate profit for shareholders, they are likely to prioritize profit over water quality if these goals conflict (Kovvali and Macey 2023). Quarterly evaluation of profits compounds this issue, incentivizing short-term financial gains over longer-term investments that would improve local infrastructure and management (Aragão 2022; Cho et al. 2023; Furlong 2010; David McDonald 2022).

Municipalities have sought to contend with these limitations with a variety of tools, including competitive bidding processes, contracting out services rather than selling services, and outsourcing to more public third parties (e.g., special districts, nonprofits). Competitive bidding introduces competition during the pre-outsourcing process because municipalities solicit proposals from all interested parties, which can lead to competitive proposals that offer higher purchase prices and commitments that align with community goals (Albalade et al. 2022; González-Gómez et al. 2011). However, once a sale or lease is approved, studies show that municipal transaction costs often increase because ensuring compliance requires significant oversight (Brown and Potoski 2003; Brown et al. 2006; Potoski et al. 2025; Xiong et al. 2022). Municipalities also outsource their utilities to public third parties, including regional special districts and non-profit firms, to reduce potential profit-seeking behavior while capturing economy-of-scale benefits (Bel and Sebó 2021; Girth et al. 2012; Mullin 2009), but public third parties are subject to similar short-term financial pressures and often mimic private-sector management practices (Aragão and Fontana 2022; Cho et al. 2023; David McDonald 2022; David. A. McDonald 2024).

Additionally, scholars have documented a recent de-privatization trend. Over the past two decades, the rate at which municipalities have privatized appears to be matched by the rate at which cities have remunicipalized (e.g., municipalities buy back utilities from private firms) (Albalade et al. 2022; David. A. McDonald 2024; Warner 2024; Warner and Aldag 2026). This countervailing trend has been especially pronounced in the water sector (Jaffer 2025; Kishimoto et al. 2015; Mann and Warner 2019; Mayol and Saussier 2024). For example, France, where more than 50% of the population is served by private water utilities, has experienced the largest share of remunicipalization (Mayol and Saussier 2024). This phenomenon has been attributed to private managers' failure to meet performance and investment goals, due in part to shareholder-value models that prioritize short-term financial returns over long-term capital investment and sustainability (Kishimoto et al. 2015; Jaffer 2025; David. A. McDonald 2024).

In sum, neither privatization nor alternative service delivery has offered a universal solution for improving service performance outcomes. Yet, despite well-documented concerns about privatization, evidence from the U.S. context complicates this critique. Several studies find that privately owned utilities exhibit stronger regulatory compliance than publicly owned systems on average (e.g., Allaire et al. 2018; Fu et al. 2020; Konisky and Teodoro 2016). Thus, private management may indeed offer

performance advantages in this sector, particularly in regulatory compliance. However, there remains a need to better align profit-seeking incentive structures with community needs. FMV legislation purports to create such alignment by restructuring acquisition and rate base approvals in ways intended to encourage consolidation of and capital investment into small, struggling utilities.

3 | Fair Market Value Legislation

FMV legislation, which shifts appraisal of utilities away from the “book value” approach, has gained traction as a potential solution to the shortcomings of traditional utility privatization (McKiernan and Warner 2024; Riley 1940). The book value approach calculates infrastructure value as the original asset investment cost (not accounting for inflation) minus depreciation. This creates a fundamental problem for both sellers and buyers. Deteriorated utilities are valued lowest precisely when they need the most investment, making transactions unattractive for both municipalities and IOUs. For example, a utility that cost \$100,000 to build in 1975 would cost approximately \$375,000 to construct today, but under book value accounting would be appraised at only \$100,000 minus fifty years of depreciation. This is akin to valuing a house at its original purchase price minus wear and tear, regardless of current market conditions. An IOU purchasing this system could only add that book valuation to its rate base as an acquisition cost, and subsequent capital improvements would also be valued under the book approach, limiting potential returns on the presumably substantial investments needed to modernize the system. Large upfront financial costs and potential political ramifications make it difficult for many municipalities to invest in their infrastructure (McKiernan and Warner 2024), perpetuating a cycle where systems deteriorate, book value valuations decline, and utility sales become financially unattractive (Bakker 2013; Grant 2013).

Proponents of state FMV legislation argue that it addresses this limitation of the book value approach (Kline 2021). Rather than valuing utility assets based solely on depreciated infrastructure, FMV statutes allow assets to be priced according to what a willing buyer would pay in an open market (McKiernan and Warner 2024). Under these laws, utility value is established through an independent appraisal—conducted by engineers and other certified appraisers—which considers factors including tangible asset condition, local market conditions, inflation, and future revenue potential (Kline 2021; McKiernan and Warner 2024). This approach often yields significantly higher valuations, especially for aged systems that still hold strategic or regional value (McKiernan and Warner 2024). Consequently, FMV laws may incentivize municipalities to voluntarily sell their water utilities because they are more likely to receive higher sale prices than would be possible under the book valuation approach.

FMV legislation also restructures the rate base approval process through which IOUs earn returns (McKiernan and Warner 2024). Privately owned utilities are regulated by state-level PUCs—technocratic boards that review utility sales and rate increases—and in states without FMV legislation, PUC rate evaluations are determined based on book valuation (as noted

above). Under FMV regulatory frameworks, PUCs are required to approve rate base increases that factor in the FMV purchase price. Additionally, these laws streamline the PUC approval process for incorporating capital investments and consolidation costs into future rate cases (GAO 2021; Kline 2021; McKiernan and Warner 2024). In turn, IOUs can pay more for distressed utilities and invest capital into these systems with confidence that they will receive a return on their investments that reflects current market conditions (Kline 2021). Therefore, FMV laws theoretically make utility sales more favorable for potential sellers and buyers.

Importantly, the presumed benefits of FMV legislation are largely predicated on the assumption that small municipalities lack the capacity to invest in their struggling water systems. These capacity constraints include limited financial resources, limited management expertise, and political barriers to raising rates (GAO 2021; McKiernan and Warner 2024). FMV laws offer municipalities the option to sell their utilities to large IOUs that can capitalize on economies of scale, dedicate professional staff and technical expertise, consolidate rates across a large consumer base, and undertake infrastructure and management upgrades efficiently and rapidly while providing municipalities with financial returns to dedicate to other priorities (Beecher 2019; GAO 2021; McKiernan and Warner 2024).

However, although potentially beneficial for municipalities and IOUs, critics argue that FMV legislation ultimately shifts the financial burden of infrastructure investment onto consumers through higher water rates (McKiernan and Warner 2024). Recent research indeed shows that private utilities tend to charge higher rates than their publicly owned counterparts, raising concerns about profit-seeking behavior and long-term affordability issues (Onda and Tewari 2021; Zhang et al. 2022; Tisdale and Suse 2022). Proponents of private ownership counter that affordability is a persistent issue in both public and private utilities and suggest it is up to states to require and offer guidance to utilities on how to improve affordability (Zhang et al. 2022). For instance, Onda and Tewari (2021) find that private utilities in California, a state with strong affordability requirements, implement more affordability programs than public utilities. Many argue further that water services in the US have long been underpriced and require billions of dollars in investment; consequently, rates must increase, whether through public or private entities (Baird 2012; Mehan 2023).

Others argue that private ownership creates an accountability concern because privately owned utilities are not overseen by local elected officials (Beecher 2019). However, state PUCs must approve utility sales, base rate increases, and capital investment commitments, as well as evaluate IOUs' returns to avoid overcharging customers (Beecher 2019; Kline 2021; Mastracchio et al. 2020; McKiernan and Warner 2024). Where IOU returns in a given year exceed allowable levels, the PUC initiates a regulatory review and either requires refunds or freezes rates until the average return on equity returns to the allowable range (Beecher 2019; Doerr 2023; GAO 2021). PUCs can also reject IOU requests for rate increases based on performance outcomes (e.g., regulatory compliance). Because FMV sales are voluntary

and IOUs face both financial and regulatory incentives to substantively improve the utilities they acquire, FMV legislation theoretically addresses many of the issues associated with privatization by rewarding capital investments and greater performance. If it is successful in doing so, the political decoupling theory suggests that FMV legislation may help align the practical considerations of privatization with electoral goals for municipal elected officials.

4 | Political Decoupling Theory and Hypotheses

Political decoupling posits that the accountability structures surrounding private and public utilities differ systematically (Teodoro et al. 2020). Municipality-owned water utilities are overseen by local elected officials, and thus, potentially unpopular policy decisions like rate increases can be attributed to those officials. Consequently, local elected officials in the U.S. often fail to increase rates out of fear of future electoral punishment (Hansen and Mullin 2022) – findings also identified in middle- and low-income countries (Budds and McGranahan 2003; Murillo 2009; Spiller and Savedoff 1999). Therefore, the potential political costs of raising rates to improve water services function as a barrier to investment in municipality-owned utilities.

Teodoro et al. (2020) argue that because U.S. privatized water utilities are not managed by local elected officials, they can take unpopular but necessary policy actions without incurring political costs. Moreover, shifting utilities to private firms may cloud blame attribution to elected officials (Barnes et al. 2025; Walker et al. 2025). The politically decoupled nature of privatized water utilities may thus provide the conditions needed to improve service. However, the key question remains: why would private utilities choose to undertake costly management improvements?

The threat of regulatory penalties is one key factor (Konisky and Teodoro 2016; Teodoro et al. 2020). Konisky and Teodoro (2016) argue that private firms are more vulnerable to financial penalties from state and federal agencies (e.g., the Illinois Environmental Protection Agency, the federal Environmental Protection Agency) and reputational damage for noncompliance. In contrast, public utilities often have more political levers to mitigate penalties politically, by arguing that limited local resources are needed for other services and that financial penalties would further strain the municipality. Private firms lack this political recourse, making them more responsive to regulatory requirements and enforcement. Similarly, Teodoro et al. (2020) show that in response to a California conservation mandate, privately owned water utilities adopted more conservation policies and conserved more water than public utilities. Moreover, PUC scrutiny and rate increase approvals are largely tied to performance outcomes, particularly regulatory compliance (Wait and Petrie 2017; Zhang et al. 2022). Together, these studies suggest that the threat of regulatory repercussions motivates privately owned water utilities to improve service delivery.

FMV legislation may also activate the potential benefits of politically decoupled service delivery. Because FMV legislation

requires PUCs to approve rate base increases that factor in both acquisition costs and capital investments, IOUs have stronger incentives to purchase and invest in struggling systems because they have greater certainty of financial returns (McKiernan and Warner 2024). The literature affirms that U.S. IOUs expect such outcomes, as they are the primary actors lobbying state legislatures to adopt FMV laws (Beecher 2019; McKiernan and Warner 2024). Additionally, unlike small municipalities with limited administrative staff, IOUs maintain professional workforces that can rapidly improve operations following acquisition (Saywitz 2026). Further, because IOUs do not face local political barriers (Teodoro et al. 2020) and are more likely to face regulatory penalties (Konisky and Teodoro 2016), we broadly expect FMV privatization to lead to improved management outcomes, particularly regarding personnel and infrastructure management.

Hypothesis 1. (Management) *FMV privatization of municipal water utilities leads to an improvement in utility management outcomes relative to utilities that remain under municipal ownership.*

We also expect FMV privatization to reduce the public's ability to link utility decision-making to local elected officials. Therefore, we expect responsibility over utility-related affairs to shift to the private owner following the utility sale, precluding electoral punishment (Teodoro et al. 2020; Walker et al. 2025). More importantly, because FMV utility sales generate significant revenue that can be invested in other local priorities, these fiscal benefits may enhance public perception of elected officials and lead voters to reward incumbents in future elections, as illustrated in the Treatment Case Qualitative Review below. Therefore, we expect FMV privatization to lead to improved electoral outcomes for local officials.

Hypothesis 2. (Political) *FMV privatization of municipal water utilities leads to an increase in incumbent elected officials' vote share relative to those in municipalities that maintain utility ownership.*

5 | Research Design

To test our hypotheses about FMV privatization, we use the Illinois case and a staggered difference-in-differences approach. In this section, we first describe Illinois's FMV regulatory framework and why Illinois provides an ideal empirical setting. We then explain our treatment and control case selection process, present a qualitative review of the treatment cases to provide context on municipality motivations and IOU investment patterns, and finally detail the data and models for our management and political analyses.

5.1 | FMV Law in Illinois

To date, at least thirteen states have adopted FMV legislation (McKiernan and Warner 2024), with many others attempting to pass similar reforms (Doerr 2020). While California adopted FMV legislation nearly thirty years ago, the other twelve states passed their legislation within the past thirteen years, with a noticeable uptick in adoption over the last decade (Kline 2021).

This study focuses on Illinois because it was one of the first states in the 21st century to adopt FMV legislation (Kline 2021) – the 2013 Illinois Water Systems Viability Act (IWSVA). Illinois is an ideal case to assess the early effects of FMV privatization in a contemporary context because it is likely more comparable to other recent adopters than California. Additionally, sufficient time has passed since the adoption of FMV legislation in Illinois for short-term local effects to materialize and be evaluated. Finally, Illinois offers an empirical advantage because the Citizens Utility Board—a nonprofit, nonpartisan organization created by the Illinois General Assembly in 1983 to advocate for consumer utility rights—has tracked every FMV-related infrastructure sale to IOUs in the state (CUB 2025), which allowed us to collect information on all cases of drinking water utility FMV privatization in the state.

The IWSVA establishes specific eligibility thresholds, appraisal procedures, and ratemaking requirements governing the acquisition process, all of which are administered and enforced by the Illinois Commerce Commission (ICC) – Illinois's PUC. Like other state FMV laws, the IWSVA allows the acquiring IOU to recover and receive a return on purchase costs and capital investments in its rate base. Importantly, rates are consolidated across communities served by the IOU in the state, helping ensure that the financial burden of acquisition and capital investment is distributed across the IOU's larger customer base.

Regarding eligibility, the acquiring IOU must serve more than 30,000 customer connections, while the utility to be sold must serve 7500 or fewer connections. For qualifying acquisitions, the ICC selects three disinterested, state-certified appraisers to independently assess the system's fair market value, with the approved value being the average of the three appraisals or the agreed-upon purchase price if it is lower than this average. Prior to submitting the final request to sell, the municipality must hold a public meeting and publicize the acquisition terms before its legislative body votes to move forward. To protect ratepayers, a single acquisition may not increase the acquiring IOU's existing customers' base rates by more than 2.5% before the next rate case held with the ICC. Post-acquisition capital investments are exempt from depreciation for ratemaking purposes for up to four years or until the next rate case, whichever comes first. The ICC determines the IOU's allowed rate of return on equity through a rate case process (typically between 8% and 10% depending on the IOU), and if actual returns exceed this threshold, the excess is returned to ratepayers through reduced future rates.

All FMV sales and associated capital investment commitments must be approved by the ICC before an IOU can include them in its rate base. This pre-approval reduces uncertainty for IOUs. In turn, FMV sale requests typically include both the acquisition cost and detailed capital investment commitments, and incentivize IOUs to begin investing immediately so those costs can be incorporated into the next rate case. Overall, the IWSVA creates aligned incentives for both sellers and buyers because municipalities have a greater likelihood of receiving a higher sale price and greater certainty that the IOU will immediately invest in the system, while IOUs gain certainty that they will receive a return on their investments.

TABLE 1 | Treatment cases.

	Treatment case	ICC sale approval	Sale price	IOU
1	Ransom	Feb. 24, 2016	\$195,208	Illinois American Water
2	Sadorus	Feb. 1, 2017	\$260,450	Illinois American Water
3	Fisher	Mar. 7, 2018	\$3,770,000	Illinois American Water
4	Farmington	Mar. 21, 2018	\$3,840,000	Illinois American Water
5	Glasford	Sep. 18, 2019	\$858,250	Illinois American Water
6	Sidney	Apr. 1, 2020	\$2,400,000	Illinois American Water
7	Leonore	Apr. 15, 2020	\$162,150	Illinois American Water
8	Andalusia	May 13, 2020	\$1,680,000	Illinois American Water
9	Jerseyville	Sep. 23, 2020	\$25,400,000	Illinois American Water
10	Livingston	Apr. 29, 2021	\$684,086	Illinois American Water
11	Mt. Pulaski	Dec. 16, 2021	\$3,510,000	Illinois American Water
12	Hardin	Apr. 7, 2022	\$2,240,000	Illinois American Water
13	Villa Grove	Sep. 15, 2022	\$7,120,000	Illinois American Water
14	Tolono	May 18, 2023	\$4,020,000	Illinois American Water
15	Broadlands	June 1, 2023	\$540,297	Illinois American Water

5.2 | Treatment and Control Case Selection Process

As of April 2025, sixteen municipal drinking water utility sales were approved by the ICC and completed following the IWSVA FMV regulatory framework. All sales were made to a single IOU: Illinois American Water Company (IAWC). IAWC is the largest investor-owned water utility in the state and serves 148 communities across thirteen operating districts or regions in the state (IAWC 2026). We include fifteen of the sixteen FMV sales for our treatment group. One case (Rosiclare, IL) was excluded because the water production and distribution systems were acquired in separate transactions multiple years apart, preventing clear identification of the acquisition date. Table 1 presents each of the treatment cases included in the study. The sale prices ranged from \$162,150 to \$25.4 million (CUB 2025).

We identified control cases using a two-stage process. First, we applied three mandatory eligibility criteria: controls had to (1) operate within the same county as a treatment case, (2) maintain municipal ownership of their water utility throughout the study period (i.e., municipalities with utilities owned by private firms, special districts, or non-profits could not serve as controls) and (3) not exceed 7500 service connections, the maximum threshold for eligible sellers under the IWSVA. Within-county matching ensures that treatment and control cases share similar regional conditions (e.g., environmental contexts, institutional arrangements, socioeconomic characteristics) and supports the use of county-level fixed effects in the regression models. Additionally, because IAWC operates larger regional systems across its thirteen operating districts statewide (IAWC 2026), a privatization occurring within a given county suggests IAWC was likely operating a larger system within that region, and small municipalities

that owned their utilities in the same county as a treated case shared a reasonably comparable acquisition likelihood.

Second, among the eligible pool of control cases within each county, we applied three selection guidelines. We prioritized including municipality controls that were closest in population size and had a similar number of service connections as the treatment cases. We used these observable characteristics because population and service connections are common proxy measures for utility operational capacity and are often associated with compliance outcomes (e.g., Allaire et al. 2018), and, importantly, population size can impact local voting behavior (Frandsen 2002; McDonnell 2020). We capped the number of controls at no more than four per treatment case to avoid over-weighting the control group in the difference-in-differences estimator, which can bias the counterfactual trend and reduce estimate precision when controls are not closely comparable to treated units. Champaign County was an exception because this county contained five treatment cases already, and within-county variation across treated municipalities itself contributes to identification. Therefore, we opted to include one control case per treatment case in Champaign. Where the eligible pool was sparser, we retained any available municipalities that met the mandatory criteria listed above to ensure each treatment case had some within-county comparison.

This matching process yielded thirty-four control cases across ten counties. Table 2 presents the full set of treatment and control cases by county and provides information on population and water utility connections for each case. As a preliminary check for comparability on these observable characteristics, we conducted two-sample *t*-tests and Wilcoxon rank-sum tests on the 2020 population, service connection count, and revenue per connection (RPC)¹ shown in Table 2. The parametric *t*-test

TABLE 2 | Descriptive characteristics of the treatment and control cases organized by county.

County	Muni.	Treatment cases				Muni.	Control cases			
		Pop. 2020	Pop. % change	Service connects	RPC		Pop. 2020	Pop. % change	Service connects	RPC
Calhoun	Hardin	801	−17.17	430	326	Batchtown	170	−20.56	160	325
						Brussels	116	−17.73	98	378
						Kampsville	310	−5.49	246	175
Champaign	Broadlands	316	−9.46	160	619	Gifford	911	−6.56	390	331
	Fisher	2062	9.62	740	416	Homer	1073	−10.06	600	173
	Sadorus	402	−3.37	192	365	Ludlow	308	−16.98	150	320
	Sidney	1208	−2.03	567	596	Ogden	729	−10.00	350	231
	Tolono	3604	4.55	1200	358	Thomasboro	1034	−8.17	490	337
Douglas	Villa Grove	2472	−2.56	1120	298	Arcola	2927	0.38	1270	429
						Atwood	1116	−8.82	600	437
						Tuscola	4636	3.48	2392	398
Fulton	Farmington	2389	−2.41	1101	268	Astoria	929	−18.58	650	511
						Cuba	1184	−8.50	625	918
						Lewistown	2041	−14.39	1280	291
Jersey	Jerseyville	8337	−1.51	4133	303	Fieldon	176	−26.36	140	450
La Salle	Leonore	121	−6.92	74	311	Cedar Point	266	−3.97	116	440
	Ransom	308	−19.79	184	179	Grand Ridge	515	−8.04	249	418
						Leland	951	−2.66	380	192
						Lostant	423	−15.06	227	436
						Naplate	412	−16.94	246	337
						Tonica	749	−2.47	324	198
Logan	Mt. Pulaski	1537	−1.85	850	240	Atlanta	1669	−1.36	768	358
						Elkhart	450	11.11	211	265
						Emden	467	−3.71	240	375
Madison	Livingston	763	−11.07	381	328	Hamel	929	13.85	394	548
						Marine	912	−5.00	431	701
						Worden	1096	4.98	470	383
Peoria	Glasford	1208	18.20	477	765	Brimfield	778	−10.37	339	563
						Dunlap	1603	15.66	541	377
						Hanna City	1253	2.29	529	819
Rock Island	Andalusia	1184	0.51	448	355	Cordova	671	−0.15	307	176
						Hampton	1779	−4.51	710	334
						Port Byron	1668	1.28	485	526
						Rapids City	964	0.52	377	374

Note: Pop. 2020 is drawn from the 2020 decennial U.S. Census. Pop. % Change is calculated from 2010 and 2020 decennial U.S. Census population counts. Service Connects reflects the number of active customer connections as of 2013 (beginning of study period), drawn from the EPA Safe Drinking Water Information System. RPC (revenue per connection) is calculated by dividing the 2012 water utility revenue by 2013 service connections; revenue data are from the U.S. Census Bureau's Census of Governments (2012).

indicates treatment municipalities are slightly larger on average (1781 vs. 1036 residents, $p=0.08$), though this difference is not significant, and the non-parametric rank-sum test is insignificant ($p=0.20$). For service connections and RPC, neither test shows significant differences between groups (service connections: $p=0.13$ and $p=0.29$; RPC: $p=0.76$ and $p=0.44$). Overall, these results suggest the treatment and control groups are reasonably comparable on observable characteristics, though treatment municipalities tend to be slightly larger, which we address in our model robustness checks. Although these tests provide confidence that our treatment and control groups are similar, a difference-in-differences design requires that treatment and control groups exhibit parallel trends in the dependent variable in the pre-treatment period. We discuss the parallel trend assumption in the management and political analysis sections below.

5.3 | Treatment Case Qualitative Review

To provide a deeper context of the fifteen FMV privatization cases in this study, we identified and assessed media reporting and press releases about each sale. Table A1 in Supplementary Appendix A shows (1) the municipality's motivation for selling its utility to IAWC, and (2) the capital investment commitments made by IAWC at the time of the sale and the progress of those commitments.

A few key themes emerge in municipality motivation. Municipality officials in IAWC press releases (Table A2) consistently mention that selling the system to IAWC would improve the community's water quality, allow for needed capital investments, and transfer management to a professional firm. Multiple cases also report that selling the utility will alleviate debt, reduce or stabilize rates, and help the municipality improve other community priorities. For example, the mayor of Mt. Pulaski stated, "Not only will residents have better quality water service from a company that has a focus on water quality, but the city will benefit from capital investments the city would not have been able to make." Collectively, the municipalities emphasize that selling the utility under the FMV framework is expected to both improve water service delivery and support other goals in the community.

We also find that fourteen of the fifteen cases included capital investment commitments from IAWC at the time of sale (Table A1), with the other case (Broadlands) indicating expected investments without a formal commitment documented at acquisition. Among cases with documented dollar amounts, commitments ranged from \$2M to \$21M to be invested over five- to seven-year periods. We found that IAWC took operational control of the water systems within weeks or months of the sale approvals by transferring day-to-day water system management from municipal staff to IAWC staff.

Upon transfer, investments began rapidly. For example, in Ransom and Sadorus, IAWC consolidated the acquired systems to the company's larger regional utility within the first year. In Glasford, water treatment plant maintenance was completed within a year. In Villa Grove, IAWC invested \$4M to replace the lead service lines within the first year. Responses from

municipalities also signal that IAWC is meeting local official goals. The mayor of Farmington stated, "I am pleased with the service Illinois American Water has provided and the genuine concern they have shown our residents. They are keeping their commitment to us and investing in our water system" (Table A2). This pace of investment aligns with our expectations of FMV legislation's incentive structures: the sooner they invest, the sooner they can begin earning returns on those investments. Overall, consistent with our expectations, municipalities chose to sell their water systems to address water quality, infrastructure, and other management challenges, while also viewing the sale as a financial opportunity to address other community priorities, and IAWC acted quickly to secure its return on investment.

5.4 | Management Analysis (H1)

To assess the managerial effects of FMV privatization, we collected data from the Safe Drinking Water Information System from 2013 through 2025 for all treatment and control cases. This database is managed by the Environmental Protection Agency and provides information on SDWA violations incurred by drinking water utilities, including the date of violation. Multiple studies have utilized SDWA violation data to assess water utility compliance and management outcomes (e.g., Allaire et al. 2018; Barnes and Farmer 2024; Konisky and Teodoro 2016; Teodoro and Switzer 2016). Water utilities (formally referred to as Community Water Systems) are cited violations for failing to comply with federal and state SDWA requirements, primarily in two categories: non-health violations and health violations. Non-health violations are issued predominantly due to utility personnel-related shortcomings (e.g., failing to maintain water quality monitoring and testing schedule, failing to report test results, failing to deliver consumer confidence reports). Health violations are issued when a utility distributes tap water containing contaminant concentrations that exceed established maximum contaminant levels or when treatment techniques do not meet minimum requirements. They are attributable to treatment technology and infrastructure shortcomings and are more directly related to human health.

Two separate managerial dependent variable measures are assessed in the management analysis and are labeled *Non-Health Violations* (i.e., count of non-health violations each month) to capture personnel-related management outcomes and *Health Violations* (i.e., count of health violations each month) to capture technical management outcomes. Three treatment utilities (Ransom, Sadorus, and Sidney) were formally consolidated into regional systems post-acquisition, at which point their individual PWSIDs (system identifiers in SDWIS) were discontinued. For the period following discontinuation in these systems, we use violations reported under the PWSID of the regional systems they were consolidated into. The unit of analysis is the drinking water utility. A reduction in violations is considered an improvement in management outcomes. Thus, we expect both *Non-Health* and *Health Violations* to decrease after FMV privatization.

Figure 1 shows the treatment timing chart for the managerial analysis and includes never-treated (no privatization),

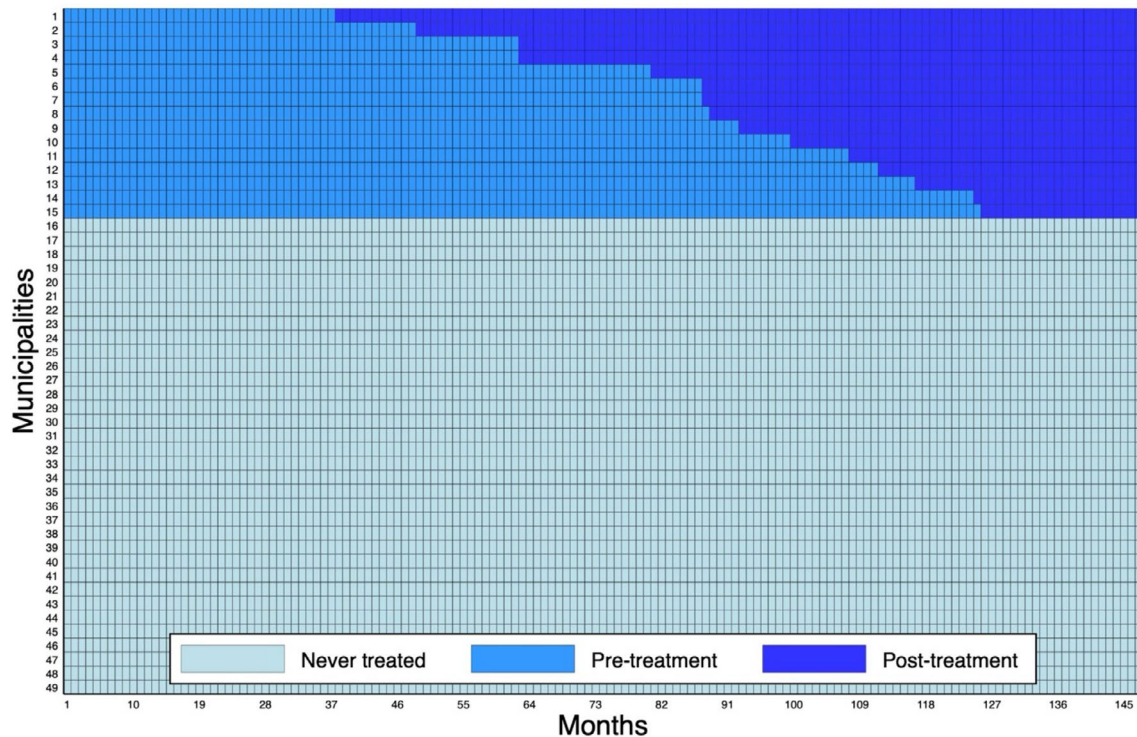


FIGURE 1 | Privatization timing chart of the managerial analysis.

pre-treatment (pre-privatization), and post-treatment (post-privatization) observations in the study. The balanced panel dataset used for this analysis includes 147 months (January 2013 through March 2025) with a total of 7203 observations.

The violation variables are count measures with a high proportion of zeros (i.e., violations rarely occur). To account for this sparsity, we estimate a two-way fixed effects (TWFE) staggered difference-in-differences (DID) model using a series of Poisson pseudo-maximum likelihood (PPML) regressions with high-dimensional fixed effects (HDFFE). We model these regressions using the *ppmlhdfe* package in STATA (Correia et al. 2020) using the following form:

$$Y_{it} = \exp(\beta_0 + \beta_1 FMV_Privatization_{it} + \alpha_i + \lambda_t + \mu_i)$$

where Y_{it} is the number of violations at utility i in month t , $FMV_Privatization_{it}$ is the treatment variable of municipality i selling their utility in month t (1 for the month the sale was approved for municipality i and all months thereafter, otherwise 0 for all pretreatment and never treated observations), α_i is a utility-level fixed effect to capture utility-level time-invariant variables, λ_t is a monthly fixed effect capturing time shocks, μ_i is county×year trend fixed effects that control for differential temporal patterns across regions (e.g., watershed quality, county level regulatory environments). Thus, β_1 provides the DID treatment effect of interest.

Recent scholarship notes that the TWFE DID modeling approach may produce a biased treatment effect in staggered settings (i.e., treated units treated at different times) if treatment changes over time and varies by individual or group. Moreover, TWFE in staggered settings can produce “forbidden

comparisons” between treated observations and earlier treated observations (Li and Strezhnev 2024). Wooldridge (2021) recommends including individual or cohort and time interactions with the treatment term to account for potential treatment variation. However, a model of this kind would require sufficient statistical power to reliably estimate dynamic treatment effects. For the managerial analysis, we have 7203 observations; however, 98% of the observations are zero for the *Non-Health Violations*, and 99% are zero for the *Health Violations*. Because the dependent variables have extremely low variation, models with multiple treatment interactions either fail to produce coefficient estimates² or are unreliable due to overfitting.

More importantly, we argue that treatment (i.e., utility sales) in this study can be considered homogenous when including municipality, time, and county×year trend fixed effects because they all (1) occurred in the same state under the same legal conditions, (2) were sales to the same IOU, and (3) occurred within a relatively short time frame (i.e., eight years). Moreover, we also expect our inclusion of many never-treated “control” cases (i.e., thirty-four municipalities) relative to the fifteen treated municipalities helps address the forbidden comparisons concern (Li and Strezhnev 2024). We also evaluate the robustness of this modeling approach by including parallel trend tests, placebo tests, and robustness checks. We recognize the limitations of using a traditional TWFE staggered DID model, but expect our approach provides the most reliable and feasible estimations possible given the data structure. All models are estimated in STATA 17 BE, and estimates are reported as incident rate ratios (IRR). Table 3 below summarizes *Non-Health Violations* and *Health Violations* among the never-treated, pre-treated, and post-treated observations.

5.5 | Political Analysis (H2)

To assess the local political effects of FMV privatization, we collected data on municipal legislators' (e.g., city council members, village trustees) election outcomes from Illinois county clerks, either from their websites or by reaching out directly to their offices via email and phone. Similar to Hansen et al. (2022), we calculated incumbent vote share for each election period in each municipality by averaging the percentage of votes incumbent elected officials received each election cycle (2015, 2017, 2019, 2021, 2023, and 2025) and exclude elections where incumbents did not run for reelection or an election wasn't held. Thus, the panel dataset for this analysis is partially unbalanced. We aggregate municipal incumbent vote shares because voters are more likely to know that their municipality's water utility was sold, but less likely to know which city council members voted to approve the sale. Municipal elections occur on April 1 of odd years in Illinois. The unit of analysis is at the municipality level, measured every odd year from April 1, 2015, through April 1, 2025 (2015 is the first election after Illinois adopted FMV legislation in May 2013). The political outcome variable is referred to as *Vote Share*. As per Hypothesis 2, we expect *Vote Share* to increase after FMV privatization.

Figure 2 shows the treatment timing chart for the political analysis across the municipalities, showing never treated (no privatization), pre-treatment (pre-privatization), and post-treatment (post-privatization) observations during the study period. Blank squares indicate that no incumbent ran for reelection or no election was held in the municipality that year. This analysis includes 226 observations.

Vote Share is a continuous measure; therefore, we estimate a TWFE staggered DID model using an ordinary least squares regression with high-dimensional fixed effects (modeled using the *reghdfe* package in STATA) (Correia 2016) using the following form:

$$Vote\ share_{it} = \beta_0 + \beta_1 FMV_Privatized_{it} + \alpha_i + \lambda_t + \mu_i + \epsilon_{it}$$

where *Vote share_{it}* is the average percentage vote share for incumbent municipal legislators at municipality *i* in election year *t*, *FMV_Privatization_{it}* is the treatment variable and takes on the value of 1 for all elections held after the utility sale was approved for a particular municipality and 0 for all pre-treatment and never treated observations, α_i is a municipality-level fixed effect to capture time-invariant factors in the community, λ_t is a

TABLE 3 | Summary of non-health and health violations.

	Never treated			Pre-treatment			Post-treatment		
	Obs.	Mean	Range	Obs.	Mean	Range	Obs.	Mean	Range
Non-Health Violations	4998	0.019	0–2	1326	0.035	0–2	879	0.002	0–1
Health Violations	4998	0.004	0–2	1326	0.020	0–2	879	0.002	0–1

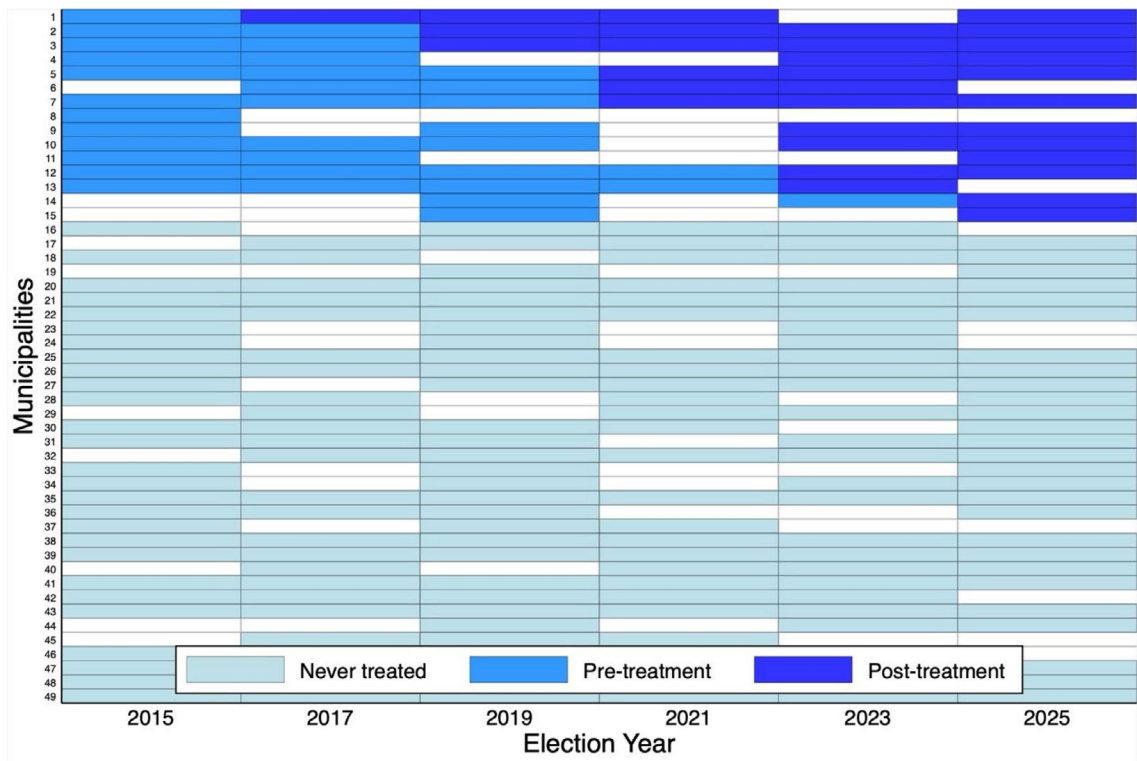


FIGURE 2 | Privatization timing chart of the political analysis.

time fixed effect capturing time shocks, μ_i is county $\times$ year trend fixed effects to capture factors impacting municipalities in the same regions, and ε_{it} is the error term. Thus, β_1 provides the DID treatment effect of interest.

Similar to the management analysis, we cannot estimate dynamic treatment effects using a more advanced staggered DID model (i.e., all time and cohort interactions) because the dataset only includes 226 observations and lacks sufficient power. Therefore, we again rely on the traditional TWFE estimation approach for the political analysis, acknowledging the potential limitations. We conduct parallel trend tests, placebo tests, and robustness checks to assess the reliability of this modeling design. All models are estimated in STATA 17 BE. Table 4 provides a summary of *Vote Share* among the never-treated, pre-treated, and post-treated observations.

6 | Results

The management and political results are presented in two separate sections below.

6.1 | Management Results

Table 5 presents the main treatment effects of FMV-related privatization on water utility SDWA compliance outcomes (reported as IRRs). In Column 1, the IRR for *FMV Privatization* is 0.026 and statistically significant at the 1% level, indicating a 97.4% reduction in non-health violations following FMV privatization. In Column 2, the IRR for *FMV Privatization* is 0.082, also statistically significant at the 1% level, indicating a 91.8% reduction in health violations after the sales. These results provide strong support for Hypothesis 1, suggesting that FMV privatization improves both personnel and technical management outcomes, relative to utilities owned by municipalities during the same period.

Table A3 provides the parallel trend tests (Supplementary Appendix B: Managerial Robustness Checks). To test the parallel trend assumption, we interacted a variable referred to as *Treat* (value of one for all cities eventually treated and a zero for all cities never treated) with a *Time* measure (month of year). Importantly, these estimations strictly excluded all post-treatment. The results show the IRRs are insignificant for both *Non-Health Violations* and *Health Violations*, suggesting violation trends between not-yet-treated and never-treated observations do not significantly vary, affirming the parallel trend assumption in the management analysis. We also estimated several robustness checks, including placebo tests, sensitivity to outlier cases, and treatment effect heterogeneity (Supplementary Appendix B: Managerial Robustness Checks), none of which meaningfully alter our results. Additionally, to assess potential outlier bias in the Poisson models, we ran binary linear probability models³ (outcome measures coded as 1 if any violation in a given month, 0 otherwise), which were also aligned with our results.

Overall, the management results suggest that FMV privatization is associated with significant reductions in both non-health and health violations, affirming Hypothesis 1. This aligns with our theoretical expectations that IOUs do not face local political barriers, are more likely to face regulatory penalties for non-compliance, and are incentivized to immediately invest in acquired systems by deploying professional staff and upgrading failing infrastructure. However, due to limitations of TWFE models in staggered treatment settings, and because we could not estimate dynamic treatment effects to produce an event study, we refrain from making causal claims. We provide more substantive discussion of these findings in the Discussion section below.

6.2 | Political Results

Table 6 presents the estimated effects of *FMV Privatization* on *Vote Share*. The results indicate that FMV privatization

TABLE 4 | Summary of vote share.

	Never treated			Pre-treatment			Post-treatment		
	Obs.	Mean	Range	Obs.	Mean	Range	Obs.	Mean	Range
<i>Vote Share</i>	160	44.5	10.8–100	34	53.9	17.8–100	32	52.5	16.9–100

TABLE 5 | Managerial DID results.

Variables	(1) Non-health violations	95% CI	(2) Health violations	95% CI
FMV privatization	0.026***	[0.005, 0.147]	0.082***	[0.012, 0.576]
Utility FE	YES		YES	
Month-year FE	YES		YES	
County $\times$ Year trend	YES		YES	
Observations	2145		476	

Note: Coefficients are reported as IRRs with 95% confidence intervals in brackets. Robust standard errors clustered at the utility level. Singleton observations automatically dropped by ppmlhdfc algorithm. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 6 | Political DID results.

Variables	(1) Vote share
<i>FMV privatization</i>	−3.964 (4.876)
Municipality FE	YES
Election year FE	YES
County × Year trend FE	YES
Observations	225

Note: Robust standard errors clustered at the municipality level. Singleton observations automatically dropped by reghdfe algorithm. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ (the effect was not significant at any of these levels).

is associated with a 3.96 percentage point decrease in incumbent vote share relative to municipalities that retained public ownership of their water utilities, but this effect is statistically insignificant. Therefore, we do not find evidence supporting Hypothesis 2. As with our management analysis, we ran parallel trend tests, placebo tests, heterogeneity tests, and robustness checks that exclude outliers, which raised no concerns regarding our results (Supplementary Appendix C: Political Robustness Checks).

Overall, we do not identify significant evidence that FMV privatization impacts incumbent elected officials' vote share, and the robustness checks affirm the study design. The null result suggests that the political mechanisms we theorized may not operate entirely as expected. First, we anticipated that selling the utility would shield elected officials from blame for future rate changes or service issues. However, voters may continue to hold local elected officials accountable, particularly when rates increase. Therefore, the positive political effects from the credit-claiming opportunities from the financial windfalls of the utility sales may be muted. Given the small sample size, we caution against drawing strong conclusions and emphasize that future work with larger samples is needed to more accurately evaluate the political consequences of FMV privatization. The next section discusses the broader implications of the management and political findings.

7 | Discussion

This study provides the first empirical assessment of FMV privatization. Guided by political decoupling theory and the broader public service outsourcing literature, we hypothesized that FMV privatization of municipal drinking water utilities would improve both utility management and political outcomes. The findings offer support for the first outcome, none for the second, and raise interesting theoretical and practical implications. Below, we discuss these implications, the key limitations, and areas for future scholarship.

Our managerial results strongly support H1 and reinforce a key tenet of the political decoupling theory: when properly motivated, privately owned utilities take actions to improve service

delivery because they do not face political costs like their public counterparts. Similar to previous studies showing private utilities outperform public utilities (Konisky and Teodoro 2016; Teodoro et al. 2020), we find that FMV privatization is associated with significant reductions in both non-health (e.g., monitoring and reporting) and health (e.g., contaminant level exceedances, poor treatment standards) violations. Therefore, our results indicate FMV privatization may activate the managerial benefits of politically decoupled utilities and may offer a valuable alternative to traditional privatization approaches, which often fail to improve public service performance (Bakker 2013; Girth et al. 2012; Warner 2024).

Importantly, FMV legislation also creates incentives that can accelerate compliance improvements. Because FMV legislation permits the state PUC to both incorporate fair market acquisition costs and subsequent capital investments into rate base approval, it creates strong incentives for IOUs to purchase and invest rapidly in struggling utilities (McKiernan and Warner 2024). Systems that require the most investment mean greater return potential, and the sooner those investments are made, the sooner they can be incorporated into base rates and start earning the IOU a return. Together, the political decoupling and FMV mechanisms provide a grounded explanation for the substantial compliance improvements in our study's treatment cases.

The management results also offer valuable substantive insights, particularly in the context of the 91.8% reduction in health violations. To contextualize these magnitudes, national SDWA compliance data reveal that approximately 6.8% of community water systems incurred health-based violations in 2021 (Allaire et al. 2018; Humphreys 2022). However, the vast majority of these violations were incurred by “small” (i.e., serve 10,000 people or less) local government-owned systems, with most occurring among systems serving 3300 people or less (Humphreys 2022). The utilities in our sample fall directly within this high-risk category. Therefore, if FMV privatization can reduce violations by over 90% in precisely the types of systems responsible for most violations nationally, this policy mechanism could offer an extremely valuable solution for reducing community exposure to waterborne contaminants and associated public health risks. Although our findings apply specifically to municipally owned water systems, not special district or county-owned systems, municipally owned systems represent a substantial share of small community water systems, making these findings directly relevant to national SDWA compliance challenges.

However, these findings come with important considerations. Governments worldwide have adopted regulatory frameworks to incentivize privatization and consolidation of water services for decades, yet growing remunicipalization trends in the 21st century raise questions about the long-term effectiveness of these governance shifts (Jaffer 2025; Mayol and Saussier 2024; Romano et al. 2023; Warner 2024). Remunicipalization has largely been attributed to private operators failing to meet long-term environmental and financial sustainability goals (Kishimoto et al. 2015; Jaffer 2025; David. A. McDonald 2024). This study measures short-term managerial effects using regulatory compliance data, but it remains unclear if FMV-authorized privatization supports longer-term community goals. Further,

Mann and Warner (2019) show that there are substantial financial and legal barriers to even large municipalities buying back utilities from IOUs. Given that FMV laws incentivize privatization among small, resource-constrained municipalities, these governments are unlikely to ever possess the financial capacity needed to buy back their utilities in the future. Therefore, understanding whether these functionally permanent governance transitions deliver long-term benefits beyond baseline regulatory compliance represents a critical priority for future FMV privatization research.

The permanence and unclear long-term impacts of FMV privatization heighten the importance of understanding the political impacts of these decisions. Our political results do not support H2. We expected that FMV privatization would improve incumbent electoral performance because political decoupling is expected to shift public scrutiny over utility affairs away from elected officials (Barnes et al. 2025; Teodoro et al. 2020; Walker et al. 2025), and elected officials could use the substantial financial windfall from FMV sales to fund other local priorities and claim credit for these actions. Thus, FMV-authorized sales were expected to enhance public perceptions of incumbents and increase the likelihood of reelection. Our review of qualitative evidence from the treated cases confirms that local elected officials planned to use the proceeds to address community needs, making the null result and negative directionality in our study puzzling. Even if selling the utility does not shift scrutiny away from elected officials, Hansen et al. (2022) find that voters do not punish local elected officials for water rate increases. Additionally, compliance outcomes improved significantly after IAWC acquired these systems, which should offset resentment of rate changes. Regardless, our findings do not indicate incumbents received any electoral benefit and may have incurred an electoral cost for selling their utilities.

One theoretical explanation for this finding is that the decision to privatize itself may spark political opposition, an argument made in both the U.S. (e.g., Mann and Warner 2019) and internationally (e.g., Hall et al. 2005). If this is indeed driving our results, it would help explain why we identify a negative, albeit insignificant, relationship and raise important questions about the political implications of FMV privatization. If these institutional arrangements generate local political pushback, it may create untenable situations for local elected officials who largely lack the means to reverse these decisions in the future. Therefore, future research with larger samples is needed to tease out the link, or lack thereof, between FMV privatization and local political outcomes.

This study has a few limitations that offer avenues for future scholarship. First, the analysis focuses on a single state, Illinois, which limits the generalizability of the findings. While Illinois was an early adopter of FMV legislation and offers a robust case for evaluating the short-term effects of FMV privatization, conditions in other states may differ from Illinois. Although we assessed nearly all municipalities that sold their drinking water utilities under Illinois's FMV regulatory framework, the managerial and political samples lacked enough variation to estimate dynamic treatment effects. Our two-way fixed effects models estimate an overall average treatment effect across all post-treatment periods; therefore, we cannot deduce how rapidly

effects emerge or whether they vary over time. This is the key reason we do not claim to have identified causal effects in this study. However, our qualitative review of each of the treatment cases suggests that most personnel-related changes and the most pressing infrastructure upgrades occurred rapidly following the utility sales, suggesting management improvements emerge within the first year or two after privatization. Given these limitations, some next logical steps for future research are to (1) collect a larger sample of FMV privatizations across multiple states to enable estimation of dynamic treatment effects and outcomes over time, and (2) compare the effects of FMV privatization to book value privatization and other forms of outsourcing (e.g., selling the utility to a regional special district) both within and across states with and without FMV laws in place.

Moreover, while assessing the effects of rate changes and issues related to affordability were outside the scope of this study, they are critical for understanding the broader implications of FMV legislation and subsequent utility sales. We encourage future scholarship to examine these variables, and particularly how FMV privatization impacts low-income households and rates. Finally, because FMV legislation represents a relatively new regulatory approach, its long-term implications, including environmental sustainability, remain unknown. Future research will be needed to evaluate whether the early management gains observed in this study persist over time and extend to other management dimensions beyond personnel and technical compliance.

8 | Conclusion

The goal of this study was to assess the managerial and political effects of FMV privatization. Most notably, we found that management outcomes—measured using personnel and technical regulatory compliance data—improved substantially after municipalities sold their water utilities to an IOU following the FMV regulatory framework in Illinois. The public health-related implications of this finding alone (i.e., 92% reduction in health violations) indicate this form of public service privatization warrants serious consideration by policy practitioners and scholars. However, the lesson from this study extends beyond water utilities.

Fundamentally, the strength of the FMV approach lies in how states can use regulatory policy to align private investor goals (profit generation) with local public service goals (capital investment, improved service delivery, financial gains for communities). If aligning private incentives with public goals can improve governance outcomes for capacity-constrained municipalities, as our findings suggest, this approach may offer valuable lessons for other infrastructure-heavy sectors facing similar challenges.

However, the effectiveness of this regulatory approach relies on state-level institutions to maintain effective oversight. In the context of FMV laws and water utilities, PUCs play a central oversight role (i.e., approving rate increases, monitoring infrastructure investments), while state environmental agencies enforce SDWA compliance requirements. The capacity and will of these institutions to maintain rigorous oversight are uncertain. A vast majority of PUC commissioners are appointed by

governors in the U.S., and a large percentage of them come from the utility industry itself, raising concerns of industry capture (Heern 2023). Other work also suggests that states and their agencies may relax environmental protection standards to support local industry (Barnes 2026; Konisky 2008). Moreover, as IOUs consolidate (e.g., the two largest water utility companies in the U.S., American Water and Essential Utilities, initiated a merger in October 2025), it may become increasingly difficult for state agencies to resist influence from these actors. In sum, regulatory frameworks like FMV that theoretically align private incentives with public goals and shift oversight to state-level institutions offer a potentially valuable governance solution for local governments. However, whether this regulatory approach produces long-term public value remains an open question.

Conflicts of Interest

The authors declare no conflicts of interest.

Endnotes

¹ We use the 2020 decennial census population because the decennial census provides more reliable population values for small municipalities than the American Community Survey, which is subject to higher margins of error, particularly for small populations. Percent population change is calculated using 2010 and 2020 decennial census counts for the same reason. Service connections are drawn from 2013, the year Illinois adopted FMV legislation, reflecting utility capacity at the start of the study period. Revenue per connection (RPC) is drawn from the 2012 Census of Governments, as this survey is conducted every five years and 2012 is the closest pre-treatment census collection year (i.e., 2017 wave postdates the first utility sale). Additionally, once a municipality sells its water utility, water utility revenue is no longer captured by the Census of Governments, as the survey collects municipal government data, and post-sale revenue accrues to IAWC as the new private owner.

² We tested more advanced staggered DID treatment approaches suggested by Wooldridge (2021) and others (Rios-Avila et al. 2021; Rios-Avila 2025), but the models failed to converge or dropped all coefficients automatically.

³ We estimate linear probability models (LPM) using `reghdfe` rather than logistic regression because logit specifications encounter separation issues with sparse binary outcomes and high-dimensional fixed effects. The `reghdfe` command efficiently absorbs high-dimensional fixed effects and handles singleton observations while providing consistent estimates of average treatment effects. Use of LPM is standard practice for panel regressions with rare-event binary outcomes (Angrist and Pischke 2009).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table A1.** Sale motivations and capital investment commitments and progress. **Table A2.** Quotes from municipal elected officials. **Table A3.** Managerial parallel trend results. **Table A4.** Management placebo test results. **Table A5.** Managerial DID result using binary outcome. **Table A6.** Political parallel trend results. **Table A7.** Political placebo test results.